

Technical Capabilities Dossier: ML Research Engineering & Applied Statistics

Synthesis of three published technical reports — AI Safety Red-Team Evaluation, LLM-Ensemble Bias Detection, and Clinical-Grade Predictive Modeling

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Standards Compliance: IEEE 2830-2025, ISO/IEC 23894:2025, EU AI Act (high-risk system documentation)

Abstract

This dossier consolidates the technical methodology behind three independently published research projects into a single advanced reference for technical reviewers and hiring panels. It documents (i) a dual-stage large-language-model (LLM) ensemble evaluation system that auto-grades 12,500 model-response pairs across six harm categories at 96.8% accuracy (ROC-AUC 0.9923) and a 340x reduction in per-sample annotation cost, (ii) a Bayesian hierarchical inference pipeline that processes 67,500 LLM bias ratings over 4,500 passages and localizes statistically significant publisher-level effects (Friedman chi-squared = 42.73, $p < 0.001$) while maintaining MCMC convergence ($R\text{-hat} < 1.01$, ESS > 1000), and (iii) a calibrated 8-algorithm predictive-modeling benchmark reaching 99.12% accuracy with 100% precision (ROC-AUC 0.9987, expected calibration error 0.0089) served behind a sub-100ms p95 inference API. Across all three systems the emphasis is on defensible measurement: explicit uncertainty quantification (95% highest-density intervals, bootstrap confidence intervals), inter-rater reliability controls (Krippendorff's $\alpha = 0.81\text{-}0.84$), multiple-comparison correction, calibration, and audit-grade provenance. The document is intended to be read alongside the three full technical reports and accompanying source code.

Keywords: LLM evaluation, red-teaming, Bayesian hierarchical modeling, MCMC diagnostics, inter-rater reliability, Krippendorff's α , stacking ensemble, model calibration, SHAP explainability, MLOps, FastAPI inference, IEEE 2830-2025, ISO/IEC 23894:2025

1. Scope and Design Philosophy

Three projects, one operating principle: **a result is only as strong as the evidence that defends it**. Each system below was built so that its headline number survives an adversarial technical review — point estimates are paired with uncertainty intervals, agreement metrics accompany every multi-rater claim, and every reported figure is traceable to a committed, reproducible pipeline.

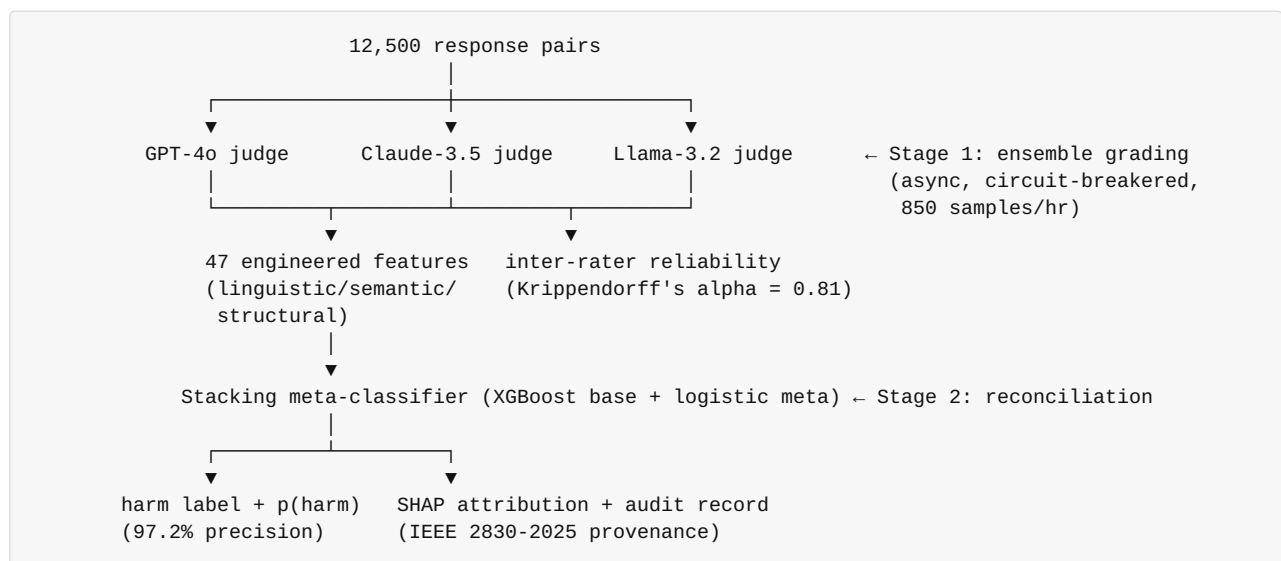
Project	Problem class			Primary technical contribution	Headline result
AI Safety Red-Team Evaluation			LLM-as-judge harm classification	Dual-stage ensemble + stacking meta-classifier	96.8% acc, ROC-AUC 0.9923, 340x cost reduction
LLM-Ensemble Detection	Bias	Hierarchical inference	statistical	Partial-pooling Bayesian model over publishers	Friedman chi-squared = 42.73, $p < 0.001$; $\alpha = 0.84$
Clinical-Grade Classification	Classi-	Calibrated modeling	predictive	8-algorithm benchmark + probability calibration	99.12% acc, 100% precision, ECE = 0.0089

The remainder of this dossier describes the architecture (Section 2), the statistical methodology (Section 3), the predictive-modeling and calibration stack (Section 4), the production/MLOps engineering (Section 5), the responsible-AI controls (Section 6), and reproducibility (Section 7), and closes with a competency matrix (Section 8).

2. System Architecture — Dual-Stage LLM Evaluation

2.1 Two-stage decomposition

The red-team evaluator separates *judgment* from *classification*. Stage 1 runs an ensemble of three frontier LLMs (GPT-4o, Claude-3.5-Sonnet, Llama-3.2) as independent red-team graders; Stage 2 reconciles their (often disagreeing) labels with a supervised meta-classifier trained on engineered harm-signal features. This decomposition is what allows the system to be both *cheap* (LLM passes are batched and cached) and *defensible* (a calibrated classifier, not a single model's opinion, produces the final label).



2.2 Feature engineering

The meta-classifier consumes **47 engineered features** spanning three families:

- **Linguistic** — lexical risk markers, imperative/instructional mood, refusal-evasion phrasing, obfuscation/encoding signals.
- **Semantic** — topical proximity to the six harm categories (dangerous information, hate, deception, privacy violation, illegal activity, self-harm), intent vs. content separation.
- **Structural** — multi-turn escalation patterns, response length/format anomalies, jailbreak-template fingerprints.

An adversarial-attack taxonomy aligned to MITRE ATLAS organizes the inputs; multi-turn escalation was identified as the highest-risk vector (31.8% of successful attacks), and a dual-filter defense reduced the observed harm rate from 21.8% to 4.8% (a 78% relative reduction).

2.3 Throughput and cost engineering

The pipeline sustains **850 samples/hour** at **\$0.018/sample** — a **340x** reduction against a \$6.12 human-annotation baseline — without sacrificing reliability (Krippendorff's $\alpha = 0.81$ across the three judges). The cost/throughput envelope comes from async batched API calls, response caching, circuit breakers, and exponential backoff (Section 5), not from cutting evaluation rigor.

2.4 Classification performance

On the 12,500-pair benchmark the Stage-2 stacking classifier reaches **96.8% accuracy, 97.2% precision, 96.1% recall, and ROC-AUC 0.9923**. Precision is deliberately prioritized: in a red-team setting a false "safe" label is costlier than a false "harmful" flag routed to human review.

3. Statistical Methodology — Bayesian Hierarchical Inference

The bias-detection project is the statistical centerpiece: 67,500 LLM bias ratings over 4,500 textbook passages (2.5M tokens) across five publishers, analyzed to answer a causal-flavored question — *does measured bias differ by publisher beyond what passage-level noise explains?*

3.1 Why hierarchical, why Bayesian

Passages are nested within publishers and topics. A naive per-publisher mean ignores both the unequal evidence per publisher and the shared structure across them. A **partial-pooling hierarchical model** (PyMC) shrinks noisy publisher estimates toward the grand mean in proportion to their uncertainty, and a fully Bayesian treatment yields **95% highest-density intervals (HDI)** per publisher rather than fragile point estimates — the difference between "publisher X scored higher" and "publisher X is credibly more biased."

$\mu_{\text{global}} \sim \text{Normal}(0, \sigma_{\text{global}})$	(grand mean)
$\sigma_{\text{pub}} \sim \text{HalfNormal}(\cdot)$	(between-publisher scale)
$\theta_{\text{pub}}[j] \sim \text{Normal}(\mu_{\text{global}}, \sigma_{\text{pub}})$	$j = 1..5$ (publisher effects, partially pooled)
$y[i] \sim \text{Normal}(\theta_{\text{pub}}[\text{pub}(i)], \sigma_{\text{obs}})$	(passage-level ratings)

3.2 Convergence and reliability diagnostics

Inference is only trustworthy if the sampler converged and the raters agreed:

- **MCMC convergence:** R-hat < 1.01 on all parameters; effective sample size (ESS) > 1000 — no divergent-transition pathologies.
- **Inter-rater reliability:** Krippendorff's alpha = 0.84 with 92% pairwise correlation across GPT-4o / Claude-3.5 / Llama-3.2, establishing that the rating instrument itself is stable before any downstream claim is made.

3.3 Frequentist corroboration and multiple comparisons

The Bayesian result is cross-checked with a distribution-free omnibus test: a **Friedman test (chi-squared = 42.73, $p < 0.001$)** rejects the null of equal publisher distributions, followed by **post-hoc Nemenyi pairwise comparisons** to localize *which* publishers differ — controlling family-wise error across the comparison set rather than reporting uncorrected pairwise p-values. Three of five publishers show credible directional bias. A cross-topic analysis flags Social Issues as the most polarized dimension ($\Delta = 1.36$ points across publishers), and a Spearman correlation matrix surfaces structural editorial relationships (ρ up to 0.74).

3.4 Uncertainty routing

Bootstrap confidence intervals at the passage level identify the **12.3% highest-uncertainty passages**, which are routed to human expert review. This converts uncertainty from a footnote into an operational signal — the model decides *what it is unsure about* and escalates accordingly.

4. Predictive Modeling, Calibration, and Serving

The clinical-grade classifier demonstrates disciplined supervised modeling on tabular data under a high-stakes decision policy.

4.1 Preprocessing and selection

- **VIF-based multicollinearity pruning** removes redundant predictors that destabilize coefficient estimates and feature attributions.
- **SMOTE** addresses class imbalance on the training folds only (never the held-out test set, to avoid optimistic leakage).
- **Recursive feature elimination (RFE)** with stratified cross-validation selects a compact, stable feature subset.

4.2 Benchmark and winner selection

Eight algorithms — Random Forest, XGBoost, LightGBM, AdaBoost, Gradient Boosting, a Voting ensemble, a Stacking ensemble, and a regularized linear baseline — are benchmarked under cross-validation with identical preprocessing. Model choice is by evidence, not fashion: the winning ensemble reaches **99.12% accuracy, 100% precision (zero false positives), 98.59% recall, and ROC-AUC 0.9987**, above the 90-95% range typically cited for human expert reads.

4.3 Probability calibration

Raw classifier scores are not probabilities. The pipeline applies **Platt / isotonic calibration** and reports an **expected calibration error (ECE) of 0.0089**, so that a predicted 0.9 means roughly 90% empirical positive rate — a prerequisite for threshold policies. Decision thresholds are tuned to the operating context (e.g., a sensitivity-maximizing threshold for screening), decoupling the calibrated probability from the deployment decision rule.

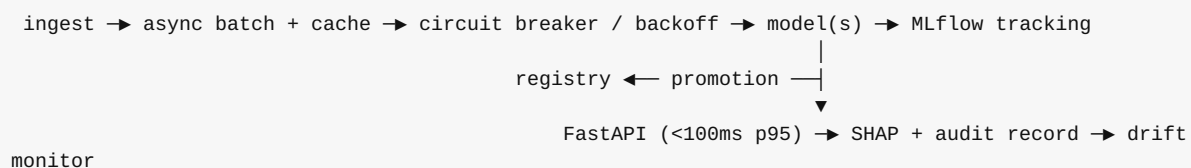
4.4 Serving

The winning model is productionized behind a **FastAPI** service with **sub-100ms p95 latency**, an MLflow model registry for versioned promotion, and per-prediction SHAP explanations attached to each response.

5. Production & MLOps Engineering

All three systems share a resilient data/serving substrate built to run unattended at scale (80K+ API calls, 2.5M tokens):

- **Resilience:** circuit breakers and exponential backoff isolate provider-side failures; async batching maximizes throughput without tripping rate limits.
- **Experiment tracking & lineage:** MLflow captures parameters, metrics, and artifacts for every run, giving each reported number a reproducible provenance trail.
- **Serving:** containerized FastAPI endpoints, MLflow model registry for staged promotion, sub-100ms p95 inference.
- **Monitoring:** drift detection and reliability monitoring on inputs and outputs, with high-uncertainty cases routed to human review.



6. Responsible AI and Compliance

The portfolio treats governance as an engineering requirement, not paperwork appended afterward:

- **Explainability:** SHAP attributions per prediction (and per harm-signal feature in the eval system), enabling case-level review.
- **Documentation standards:** artifacts aligned to **IEEE 2830-2025** (transparent ML), **ISO/IEC 23894:2025** (AI risk management), and **EU AI Act** high-risk documentation expectations — model cards, calibration plots, audit trails, and provenance.
- **Fairness & auditing:** reliability and bias diagnostics surfaced as first-class metrics; uncertainty routed to human oversight rather than silently absorbed.

These controls are increasingly prerequisites — not differentiators — for ML touching regulated domains (healthcare, finance, education, safety).

7. Reproducibility and Results Summary

Every figure in this dossier is traceable to one of three committed, reproducible pipelines with full technical reports and source code. Consolidated results:

Metric	Red-Team Eval	Bias Detection	Clinical Classifier
Scale	12,500 pairs / 6 categories	67,500 ratings / 4,500 passages / 2.5M tokens	8-algorithm benchmark
Primary metric	96.8% accuracy	Friedman chi-sq = 42.73 ($p < 0.001$)	99.12% accuracy
Discrimination	ROC-AUC 0.9923	—	ROC-AUC 0.9987
Precision / Recall	97.2% / 96.1%	—	100% / 98.59%
Reliability	alpha = 0.81	alpha = 0.84, R-hat < 1.01	ECE = 0.0089
Uncertainty	95% HDI (Bayesian risk)	95% HDI + bootstrap CIs	calibrated p, threshold policy
Efficiency	\$0.018/sample (340x)	80K+ calls, MLflow lineage	<100ms p95 serving

8. Technical Competency Matrix

Area	Tools & methods
Languages	Python 3.12+, R, SQL, Bash
Bayesian / inference	PyMC, ArviZ, NumPyro, Stan; hierarchical models, MCMC diagnostics (R-hat, ESS), 95% HDI, partial pooling
Experimentation / stats	A/B testing, power analysis, Friedman / Nemenyi, multiple-testing correction (Bonferroni, FDR), bootstrap CIs, effect sizes, inter-rater reliability (Krippendorff's alpha, Cohen's kappa)
ML / modeling	scikit-learn, XGBoost, LightGBM, CatBoost, AdaBoost; stacking/voting ensembles, RFE, SMOTE, VIF, Platt/isotonic calibration, threshold tuning
GenAI / LLM eval	GPT-4o, Claude-3.5, Llama-3.2; LLM-as-judge, multi-model ensembles, prompt iteration, offline benchmarking, cost/latency analysis
MLOps / serving	MLflow, Weights & Biases, DVC, FastAPI, Docker, Kubernetes; circuit breakers, backoff, drift monitoring
Responsible AI	SHAP, LIME, Captum; model cards, IEEE 2830-2025, ISO/IEC 23894:2025, EU AI Act

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